

Use of Immune Checkpoint Inhibitors in the Treatment of High-Risk, Early-Stage Triple-Negative Breast Cancer: ASCO Guideline Rapid Recommendation Update Q and A

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Triple-negative breast cancer (TNBC) is associated with a higher likelihood of distant recurrence and death compared with other subtypes of breast cancer.¹ The use of anthracycline- and taxane-based chemotherapy regimens in the adjuvant setting is standard for high-risk TNBC. Recently, several studies have shown that the addition of immune checkpoint inhibitors improves pathologic complete response (pCR) in TNBC, and selected trials now also show improved survival outcomes.

In February 2022, Schmid et al² reported the results of the fourth interim analysis of data from the KEYNOTE-522 trial. The trial evaluated the addition of pembrolizumab (n = 784) versus placebo (n = 390) to neoadjuvant chemotherapy (carboplatin and paclitaxel followed by doxorubicin or epirubicin and cyclophosphamide) with completion of 1 year of pembrolizumab or placebo after surgery, in 1,174 patients with previously untreated, stage II or stage III TNBC. Event-free survival (EFS) was superior among patients who received pembrolizumab compared with patients who received placebo.

ASCO published a rapid recommendation update of ASCO's guideline on neoadjuvant chemotherapy, endocrine therapy, and targeted therapy for breast cancer (Fig 1) to address the role of pembrolizumab in the treatment of high-risk, early-stage TNBC.^{3,4} This companion Q and A article addresses some of the questions that clinicians may encounter as they incorporate these recommendations into clinical practice.

QUESTION: SHOULD PEMBROLIZUMAB BE CONTINUED IN PATIENTS WHO ACHIEVE pCR AFTER NEOADJUVANT CHEMOTHERAPY PLUS PEMBROLIZUMAB?

The phase III KEYNOTE-522 trial included continuation of pembrolizumab for nine cycles (27 weeks) after surgery in adjuvant setting, regardless of pathologic response to neoadjuvant therapy. Although the subgroup analysis showed more pronounced absolute

EFS benefit in patients who failed to achieve pCR (67% with pembrolizumab v 56% with placebo) versus those who achieved pCR EFS (94% v 92%), we support continuation of pembrolizumab in the adjuvant setting in all patients while we await data from other trials addressing this question.

QUESTION: CAN A CHEMOTHERAPY REGIMEN DIFFERENT FROM THE ONE USED IN KEYNOTE-522 BE USED WITH PEMBROLIZUMAB?

It is difficult to estimate the relative contribution of the neoadjuvant chemotherapy regimen used in KEYNOTE-522 toward the improved pCR and EFS. Although data on the benefit of carboplatin in the neoadjuvant setting are conflicting,⁵⁻⁷ in the absence of a definitive phase III trial showing improvement of outcomes when combining checkpoint blockade with other neoadjuvant regimens, we support the use of the full KEYNOTE-522 regimen. However, if a patient experiences toxicity, it is not unreasonable to dose reduce or discontinue the drug. In patients with TNBC who have contraindications to anthracycline therapy and are being considered for regimens such as docetaxel and cyclophosphamide, it would be reasonable to add pembrolizumab to their regimen.

QUESTION: SHOULD ADJUVANT CAPECITABINE BE ADMINISTERED WITH PEMBROLIZUMAB IN PATIENTS WHO FAIL TO ACHIEVE pCR WITH PEMBROLIZUMAB-BASED NEOADJUVANT THERAPY?

Patients with TNBC who have residual disease after neoadjuvant chemotherapy are currently offered adjuvant capecitabine chemotherapy on the basis of improved survival shown in CREATE-X trial.⁶ It should be noted that capecitabine was not permitted in KEYNOTE-522; however, there are safety data for the capecitabine-pembrolizumab combination in the advanced setting.⁷ Therefore, it is reasonable to administer capecitabine concurrently or sequentially in patients at high risk of recurrence. The long-term safety of this combination is not known.

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Recommendation	Type	Evidence quality	Strength
3.4. For patients with T1cN1-2 or T2-4N0 (stage II or III), early-stage triple-negative breast cancer, the Panel recommends use of pembrolizumab (200 mg every 3 weeks or 400 mg every 6 weeks) in combination with neoadjuvant chemotherapy, followed by adjuvant pembrolizumab after surgery. Adjuvant pembrolizumab may be given either concurrent with or after completion of radiation therapy. Given that immune-related adverse events (irAEs) associated with pembrolizumab therapy can be severe and permanent, careful screening for and management of common toxicities are required. The ASCO guideline for management of irAEs in patients treated with immune checkpoint inhibitor therapy offers detailed practice recommendations and should be consulted by clinicians who prescribe pembrolizumab for patients with early-stage TNBC, https://ascopubs.org/doi/full/10.1200/JCO.21.01440. <i>Qualifying statements: The results from KEYNOTE-522 are based on continued pembrolizumab in the adjuvant setting. There is uncertainty concerning the optimal adjuvant treatment, given independent benefits of capecitabine in TNBC and olaparib in patients with germline BRCA mutations without pembrolizumab. There are no data to support the use of pembrolizumab in combination with either capecitabine or olaparib.</i>	EB	M	M

FIG 1. 2022 updated recommendation. EB, evidence-based; M, moderate; TNBC, triple-negative breast cancer.

QUESTION: SHOULD OLAPARIB BE ADDED TO PEMBROLIZUMAB FOR gBRCA1m AND/OR gBRCA2m CARRIERS WITH TNBC WHO HAVE RESIDUAL DISEASE AFTER NEOADJUVANT CHEMOTHERAPY?

Patients with TNBC are more likely to harbor a gBRCA1m and/or gBRCA2m, and such mutations have been reported in about 15% of patients with TNBC.⁸ The OlympiA trial demonstrated a significant improvement in 3-year invasive disease-free survival (DFS; 85.9% v 77.1%, hazard ratio 0.58, $P < .001$) and distant DFS (87.5% v 80.4%, $P < .001$) with the addition of 1 year of adjuvant olaparib, a poly (ADP-ribose) polymerase inhibitor, to standard therapy in gBRCA1m and/or gBRCA2m carriers with high-risk early-stage human epidermal growth factor receptor 2–negative breast cancer.⁹ KEYNOTE-522 did not select for gBRCA1m and/or gBRCA2m, and olaparib was not permitted. Data in the setting of advanced disease support the safety of combining checkpoint point blockade and poly (ADP-ribose) polymerase inhibitors.¹⁰ Therefore, the use of adjuvant pembrolizumab and olaparib concurrently or sequentially can be considered for eligible patients. The relative risks and benefits of a concurrent approach should be weighed, as the long-term safety of these combinations is not known.

QUESTION: SHOULD OTHER CHECKPOINT INHIBITORS BE USED IN THE NEOADJUVANT SETTING INSTEAD OF PEMBROLIZUMAB?

Other immune checkpoint inhibitors have been tested in the neoadjuvant setting among patients with TNBC. Impassion 031¹¹ showed improved pCR with the addition of atezolizumab to neoadjuvant chemotherapy, but the study was not powered to evaluate EFS. Durvalumab was tested in the phase II GeparNuevo study¹² and showed increased pCR and improved invasive DFS and OS; however, the trial was not powered for evaluation of long-term outcomes. As only pembrolizumab has received regulatory approval, we do not recommend use of alternate immunotherapeutic agents.

QUESTION: SHOULD ADJUVANT PEMBROLIZUMAB BE USED IN PATIENTS WITH RESIDUAL DISEASE AFTER NEOADJUVANT CHEMOTHERAPY WITHOUT PEMBROLIZUMAB?

There are currently no data to support adjuvant pembrolizumab use in patients who did not receive neoadjuvant pembrolizumab. Ongoing trials, including SWOG S1418, will clarify this issue.

In summary, addition of pembrolizumab to chemotherapy in the neoadjuvant setting followed by continuation in the adjuvant setting is the new standard of care for patients with high-risk TNBC as defined in KEYNOTE-522. Further

research and follow-up data are needed to determine whether adjuvant pembrolizumab is needed for patients

who achieve complete pCR. Ongoing trials will guide use of other immune checkpoint inhibitors.

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DISCLAIMER

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Use of Immune Checkpoint Inhibitors in the Treatment of High-Risk, Early-Stage Triple-Negative Breast Cancer: ASCO Guideline Rapid Recommendation Update Q and A**

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